Website Traffic Analysis Report

This project focused on analyzing website traffic data from the Kaggle dataset “Website Traffic Analysis” to understand user behavior, session patterns, landing pages, referral sources, and overall engagement. The dataset was cleaned and processed using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn. Missing values, duplicate records, and inconsistent formats were handled to ensure accurate analysis and reliable metric calculations.

The analysis measured important website performance metrics including total users, total sessions, bounce rate, average session duration, top landing pages, top exit pages, and referral traffic sources. Session-based analysis helped identify how users navigated through the website and which pages contributed most to engagement or drop-offs. User flow visualization revealed that a significant portion of visitors entered through a few high-traffic landing pages, while several exit pages showed higher abandonment rates, indicating possible usability or content issues.

Referral source analysis showed that search engines and direct traffic generated the majority of website visits, while social media and external referral platforms contributed a smaller percentage of traffic. Bounce rate analysis indicated that some landing pages were not effectively retaining users, suggesting a mismatch between visitor expectations and page content. Average session duration also varied across traffic sources, showing that users arriving from organic search spent more time exploring the website compared to visitors from other channels.

Based on the findings, several optimization strategies are recommended for Alfido Tech. First, improve the design and content quality of high-bounce landing pages to increase engagement. Second, optimize page loading speed to reduce user drop-offs and improve user experience. Third, strengthen internal linking and navigation structure to guide users toward important conversion pages. Fourth, enhance SEO and content marketing strategies to attract more high-quality organic traffic. Finally, implement targeted call-to-action sections and personalized recommendations to improve user interaction and conversion rates.

Overall, the project successfully demonstrated how website traffic analysis can help organizations understand user behavior, identify weak points in the customer journey, and make data-driven decisions to improve website performance and conversions.